Introduction

Cold neutral atoms are a versatile platform for quantum simulation, precision measurements, and quantum information processing. This thesis concerns the development of an experiment at the University of Bologna aimed at preparing, controlling, and eventually transporting a gas of laser-cooled rubidium-87 (^{87}Rb) atoms into a Hollow-Core Photonic Crystal Fiber (HCPCF).

Unlike conventional free-space optical traps, where the interaction volume is limited by diffraction, hollow-core fibers allow atoms to interact with a guided optical mode over extended propagation distances [1]. This configuration is relevant to applications requiring long interaction lengths, such as matter-wave interferometry [2], quantum memories, and the investigation of collective phenomena[3, 1].

A central challenge is transferring cold atoms from a free-space trap into the hollow core while preserving a low motional energy. The experimental sequence considered here relies on cooling atoms in a Magneto-Optical Trap (MOT) and optical molasses, loading them into an Optical Dipole Trap (ODT), and transporting them via an optical conveyor belt formed by two counter-propagating 1064 nm fields.

This thesis focuses on the experimental and numerical steps required to establish these capabilities. The work includes optimizing the optical molasses, characterizing the ODT loading, and implementing the second optical branch for the conveyor belt. Furthermore, since the transport process can increase the motional energy of the atoms, a multilevel numerical model of Electromagnetically Induced Transparency (EIT) cooling is developed to investigate the atomic dynamics and the restoration of a low-entropy state.

Chapter 1 introduces the theoretical concepts covering HCPCFs, laser cooling, optical trapping, and EIT cooling. Chapter 2 describes the experimental apparatus, the calibration and measurement procedures, the fiber-injection system design, and the formulation of the numerical model. Chapter 3 presents the experimental results regarding molasses optimization, ODT capture efficiency, and field stability, alongside the simulated EIT cooling dynamics.

The results presented here establish the experimental infrastructure and the numerical framework required for a future implementation of controlled atom transport and cooling inside the HCPCF. The complete conveyor-belt sequence and its experimental validation inside the fiber constitute the next stage of the experiment.